

Yash Jain

Austin, TX

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Education

University of Pennsylvania

PhD. Computer and Information Science

Philadelphia, PA

August 2026 - May 2030

The University of Texas at Austin

B.S. Computational Engineering, Honors - GPA: 3.97/4.00

Austin, TX

August 2023 - May 2026

Distinguished College Scholar. Relevant Coursework: Scientific Computation, CFD, Structural Analysis, Linear Systems, Software Engineering, Engineering Computation. Planned Coursework: Senior Design, Applied Time Series, Electromechanical Systems

The University of Texas at Austin

B.S. Mathematics - GPA: 3.97/4.00

Austin, TX

August 2023 - May 2026

Distinguished College Scholar. Relevant Coursework: Real Analysis, Algebra, Numerical Linear Algebra, Probability & Statistics, Mathematical Modeling. Planned Coursework: Partial Differential Equations, Numerical Differential Equations, Mathematical Modeling

Research Experience

NomadicML - Control and Learning for Autonomous Robotics Lab

Austin, TX

Researcher, Out-of-Distribution (OOD) Detection

Jan 2026 - Present

- Collaborated with NomadicML on an industry-focused research initiative to develop anomaly detection and reliability evaluation methods for video foundation models used in autonomous driving and physical AI applications.
- Developed large-scale video anomaly detection pipelines using NVIDIA Cosmos Predict 2.5, VideoMAE, V-JEPA, and other state of the art foundation models to identify safety-critical and out-of-distribution driving events across approximately 120,000 frames from real-world driving datasets.
- Developed novel evaluation frameworks based on SCOPED score statistics, kernel density estimation, and diffusion latent representations, improving anomaly separation performance by 20% under distributional shift conditions.
- Scaled experimentation through GPU-accelerated PyTorch pipelines, latent caching strategies, and AzureML batch inference workflows, enabling automated benchmarking across 200+ simulations and multiple datasets including HDD, UCF-Crime, MNIST, CIFAR-10, CIFAR-100, and CelebA.

Control and Learning for Autonomous Robotics Lab - Dr. David Fridovich-Keil

Austin, TX

Researcher, Multi-Agent Game Theory for Autonomous Driving

Jan 2025 - May 2026

- Built a Julia/Python framework for multi-agent dynamic games, integrating MCP solvers, receding-horizon control, and CARLA simulation for real time driving scenarios.
- Developed a multimodal intent inference system combining inverse game solving, VAEs, and DINO embeddings; improved goal prediction stability across ambiguous rollouts.
- Created differentiable simulation modules to enable gradient based optimization over multi-agent interactions.
- Targeting a journal submission to IJRR imminently.

Moncrief Summer Research Internship, Oden Institute

Austin, TX

Research Intern

May 2025 - Aug 2025

- 6,500 Stipend + Housing/Meals/Travel
- Investigated multimodal driver intent inference using DINO vision models, LSTM based encoders, and latent space alignment via VAEs.
- Designed cross modal fusion networks that improved planning robustness in multi agent settings under sensor noise and occlusion.
- Improved multimodal inference pipeline by refining latent space alignment techniques and validating robustness across diverse driving rollouts.
- Presented a research poster to 30+ faculty, postdocs, and industry researchers at the Moncrief Summer Scholar's Poster Session.

Texas Spacecraft Laboratory - Dr. Brandon Jones

Austin, TX

Command & Data Handling Engineer, SCOPE-1 Mission

Aug 2024 - Oct 2025

- Developed satellite flight software in NASA F' on BeagleBone Black, handling telemetry, fault protection, and command execution.
- Integrated subsystem data streams into a robust CDH pipeline; validated performance through unit tests, hardware in loop integration, and orbital simulation.
- Debugged and resolved embedded software failures across flight systems during integrated testing.

Longhorn Rocketry Association

Austin, TX

Payload Engineer

Aug 2024 - May 2025

- Designed and 3D printed payload components in SolidWorks for launch vehicle integration; collaborated on structural layout and sensor housing.
- Integrated oxygen production experiments with sensor arrays; calibrated for in flight data capture and real time telemetry.
- Led end to end system integration and version control across payload revisions, reducing integration time.

Texas Spacecraft Laboratory - Dr. Brandon Jones

Austin, TX

Software Engineer, High-Altitude Weather Balloon Mission

Aug 2023 - Aug 2024

- Developed an embedded avionics stack using Arduino and custom PCBs for telemetry and data logging in extreme conditions.
- Implemented fault tolerant sensor interfacing and mission critical firmware for low pressure, sub zero environments.
- Improved flight data integrity through redesign of software-hardware integration and telemetry buffering.

Teaching & Academic Service

Computational Engineering Department, UT Austin

Austin, TX

Tutor Manager

Aug 2024 - May 2026

- Delivered 100+ one on one tutoring sessions for core courses in computation, statics/dynamics, and software engineering.
- Improved student outcomes by scaffolding complex concepts through interactive problem breakdowns and visual modeling.

Department of Physics, UT Austin

Austin, TX

Learning Assistant, PHY 329C: Computational Physics

Aug 2025 - Dec 2025

- Guided 30+ students in topics including numerical linear algebra, ODE/PDE solvers, optimization, and scientific Python.
- Applied active learning and cognitive science strategies to deepen conceptual understanding and foster student led discovery.

Sanger Learning Center, UT Austin

Austin, TX

Teaching Assistant, M 110T: Business Calculus

Jan 2025 - May 2025

- Led weekly discussions and exam reviews for 10+ students, supporting applied calculus learning in a business context.
- Created worksheets and visual aids linking calculus to optimization, marginal analysis, and real world business models.

Sanger Learning Center, UT Austin

Austin, TX

Peer Tutor (Math, Physics, Engineering)

Aug 2024 - Aug 2025

- Conducted 200+ tutoring sessions with a 5.0/5.0 rating, supporting student success in calculus, physics, programming, and DEs.
- Designed training guides, peer interview rubrics, and academic workshops that improved tutor preparation and consistency center wide.

Leadership & Community

Texas Spacecraft Laboratory

Austin, TX

Student Director

Aug 2025 - May 2026

- Led operations for a 100+ member research lab, managing logistics, access control, onboarding, and equipment scheduling.
- Designed the Freshman Exploratory Initiative, enabling 25+ first year students to contribute to satellite systems via guided systems engineering curriculum.
- Oversaw Class-100 cleanroom installation by coordinating vendors, procurement, and facilities for mission ready integration.

Texas Spacecraft Laboratory

Austin, TX

Facilities Manager

Nov 2023 - Aug 2025

- Managed thermal/vacuum chambers and additive manufacturing tools, supporting spacecraft testing and development.
- Implemented a lab wide equipment check out and point of contact system, reducing delays and increasing lab utilization.
- Spearheaded deployment of a new industrial 3D printer and created onboarding workflows for multiteam usage.

AIAA, UT Austin

Austin, TX

Treasurer

Aug 2023 - May 2025

- Managed finances for a 100+ member student organization supporting technical competitions, outreach, and national convention participation.
- Increased organizational budget by 30% through new industry sponsorships, profit sharing events, and merchandise expansion.

Association of Computational Engineering (ACE)

Austin, TX

Treasurer

Aug 2023 - May 2025

- Oversaw budget for departmental org representing 30–40% of Computational Engineering majors; supported hackathons, workshops, and speaker events.
- Led planning and execution of major initiatives including coding competitions and technical interview prep series.

Publications & Presentations

2025: Y. Jain, X. Liu, L. Peters, “Bayesian Inverse Games with High-Dimensional Multi-Modal Observations,” under review at *International Journal of Robotics Research (IJRR)*.

2025: *Learning To Drive: Vision Meets Game Theory*. Poster presented at the Moncrief Summer Scholar’s Poster Session, Oden Institute.

2024: Contributor to PutnamBench, an open-source benchmark for automated theorem proving, presented at the ICML AI4MATH Workshop (2024). Contributed implementation; not a publication author.

Skills

Languages: Python, Julia, C++, MATLAB, Java, Rust

ML & AI: VAEs, LSTMs, Vision Transformers, multimodal fusion

Control & Planning: MCP solvers, receding-horizon control, Nash computation, intent inference

Simulation & Autonomy: CARLA, trajectory optimization, real time integration, differentiable simulation

Numerical Methods: ODE/PDE solvers, nonlinear optimization, constraint systems

Tools & Platforms: ROS2, Git, Docker, Linux, SolidWorks